

Problem 1

Let $S_n = X_1 + \cdots + X_n$

$$\begin{aligned}
 \varphi_{S_n/n}(t) &= E(e^{itS_n/n}) = \prod_{k=1}^n E(e^{itX_k/n}) \\
 &= \prod_{k=1}^n \varphi_{X_k}(t/n) \\
 &= \prod_{k=1}^n \left(1 + \frac{it\mu}{n} + O\left(\frac{\text{Var}(X_k)t^2}{n^2}\right)\right) \\
 &= \prod_{k=1}^n \left(1 + O\left(\frac{\text{Var}(X_k)t^2}{n^2}\right)\right)
 \end{aligned}$$

Then take limit as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \varphi_{S_n/n}(t) = \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + O\left(\frac{\text{Var}(X_k)t^2}{n^2}\right)\right) = 1 = e^0$$

Since $O\left(\frac{\text{Var}(X_k)t^2}{n^2}\right)$ goes to zero

Thus $S_n/n \xrightarrow{(d)} 0$

Problem 2

Shown in class $\varphi_X(t) = e^{-t^2/2}$ and $f_X(t) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$

Thus $\varphi_{XY}(t) = E(e^{itXY}) = E(E_Y(e^{itXY}|X))$ thus since they are independent $E_Y(e^{itXY}|X) = \varphi_Y(tX)$
 $E(\varphi_Y(Xt)) = E(e^{-t^2X^2/2})$

$$\begin{aligned} E(e^{-t^2X^2/2}) &= \int_{-\infty}^{\infty} e^{-t^2x^2/2} f_X(x) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2x^2/2} e^{-x^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2x^2(1+t^2)} dx \\ &= \frac{1}{\sqrt{2\pi}} \sqrt{\frac{2\pi}{1+t^2}} \\ &= \frac{1}{\sqrt{1+t^2}} \end{aligned}$$

Thus for the next part since they are all independent we can just use the function we found in the first part then multiply them together thus we get $\varphi(t) = \frac{1}{1+t^2}$

$$\begin{aligned} \varphi_X(t) = E(e^{itX}) &= \int_{-\infty}^{\infty} e^{itx} (1/2) e^{-|x|} dx \\ &= (1/2) \int_{-\infty}^0 e^{itx} e^x dx + (1/2) \int_0^{\infty} e^{itx} e^{-x} dx \\ &= \frac{1}{2(it-1)} + \frac{1}{2(1+it)} \\ &= \frac{1}{1+t^2} \end{aligned}$$

Thus since the characteristic of $f(x) = (1/2)e^{-|x|}$ is the same we know that the pdf of $X_1X_2 + X_3X_4$ is $(1/2)e^{-|x|}$

Problem 3

$$\begin{aligned}
\varphi_Z(t) &= E(e^{itZ}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{itz} f(x, z + 2x) dx dz \\
&= (1/2\pi) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{itz} \exp\left(-\frac{5x^2 - 4xz - 8x^2 + z^2 + 4xz + 4x^2}{2}\right) dx dz \\
&= (1/2\pi) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{itz} e^{-(1/2)(x^2 + z^2)} dx dz \\
&= (1/2\pi) \int_{-\infty}^{\infty} e^{itz - (1/2)z^2} \int_{-\infty}^{\infty} e^{-(1/2)x^2} dx dz \\
&= (1/2\pi) \frac{\sqrt{\pi}}{\sqrt{1/2}} \int_{-\infty}^{\infty} e^{-(-itz + (1/2)z^2)} dz \\
&= (1/2\pi) \frac{\sqrt{\pi}}{\sqrt{1/2}} \frac{\sqrt{\pi}}{\sqrt{1/2}} e^{-t^2/2} \\
&= e^{-(1/2)t^2}
\end{aligned}$$

Thus we know that Z is an $N(0,1)$ since it has the same character.